Data-Backed Journal-Style Replication of *Lens or Prism? Patent Citations as a Measure of Knowledge Flows from Public Research*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Lens or Prism? Patent Citations as a Measure of Knowledge Flows from Public Research using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 2,511 observations across 178 entities, and the batch headline estimate is 0.0220 for External Public-Research Citations with p = 0.59. The local panel treats external public-research citations as the clearest executable public-research channel and compares them directly against an internal citation channel. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: public research; patent citations; nonpatent references; knowledge flows; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Lens or Prism? Patent Citations as a Measure of Knowledge Flows from Public Research and treats the source article as an empirical argument that

has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether citation-based indicators capture knowledge flows from public research or distort them through measurement error (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because citation measures are used constantly in strategy research even when the knowledge-flow mechanism itself is unobserved. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 2,511 observations across 178 entities and spans 1990 to 2014 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Roach and Cohen, 2013; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent local measurement analogue that compares public-research-oriented citation channels inside a current executable firm-year panel. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Roach and Cohen, 2013; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal literature on patents as indicators, knowledge flows from science, and measurement error in innovation research. (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is fundamentally about whether an observable citation channel is informative enough to stand in for an unobserved knowledge-flow process. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the

source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The review therefore adds classic references on patent indicators, spillovers, and the use of citation-based proxies in strategy research. (Jaffe, 1986; Hall et al., 2005; Griliches, 1990). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Fleming and Sorenson, 2004; Rosenkopf and Almeida, 2003).

2 Original Measurement Logic and Replication Boundary

The focal source article is (Roach and Cohen, 2013). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper evaluates whether citations are valid measures of knowledge flows from public research. The central empirical contrast is between alternative citation channels. The paper concludes that nonpatent references are often better measures of public-research knowledge flows than patent references. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal

is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Roach and Cohen (2013) starts with the published empirical design rather than with the local rerun. The source article builds a newly constructed dataset that matches patents to survey responses at the research-and-development-lab level and uses that match to validate citation-based measures of knowledge flows from public research (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published sample matches patents to research-and-development labs and survey-based measures of knowledge flows from public research. The published measurement architecture is described as follows: The published design compares survey-based knowledge-flow measures with backward patent and nonpatent reference channels. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: the paper's lab-linked patent analysis comparing survey-based knowledge-flow measures with backward patent and nonpatent reference channels, together with an illustrative measurement-error exercise for innovative-performance regressions. The key empirical task is measurement validation, not a generic productivity regression. The article asks whether alternative citation channels recover the same public-research knowledge flows observed in the survey-linked lab data. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is a methodology-specific local analogue because the original R&D-lab survey match is not part of the current

processed-data stack. The local manuscript can evaluate the public-research citation logic only through firm-year science and citation files, so it cannot claim a direct replay of the source validation design (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper is about measuring knowledge flows from public research. The local reconstruction therefore adopts the following bounded decision: The local rerun uses external public-research citations as the public-research channel observable in the workspace. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Alternative citation channels are compared directly. The local reconstruction therefore adopts the following bounded decision: Internal science citations provide the comparison channel in the executable panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Measurement validity rather than pure production effects is the center of the argument. The local reconstruction therefore adopts the following bounded decision: The outcome is next-year patenting, which makes the local paper a predictive measurement analogue rather than

a direct survey-validation replay. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Citation Channels, and Sample

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Firm-year publication, patenting, and corporate NPL citation files already merged into the executable science panel. External public-research citation counts and internal science-channel counts from the local DISCERN-linked firm-year data. Compustat-linked scale and performance controls already available in the workspace (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 2,511 observations across 178 entities and covers 1990 to 2014 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue uses a firm-year panel with external public-research citation counts, internal science-channel citations, and publication stocks to evaluate which observable

channel is most strongly tied to subsequent patenting. The local sample frame is DISCERN-linked public-firm year panel with publications, patenting, corporate NPL citation counts, and financial controls The observed time window in the executable panel is 1990 to 2014 in the local executable panel The outcome is Next-year patent count at the firm-year level (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Roach and Cohen, 2013; Hall et al., 2005).

3.1 Local Dataset Families Used in the Measurement Analogue

This replication uses three verified local dataset families already present in the workspace: the DISCERN-linked firm-year science panel, the processed patent and nonpatent citation panel that records external public-research and internal science-channel use, and Compustat-linked firm controls (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: discern-linked firm-year science panel is used to define the firm-year sample and publication stock; processed patent and nonpatent citation panel is used to measure external public-research citations and internal science-channel citations; compustat-linked firm controls is used to supply scale, profitability, and r&d controls. Together, they support 2,511 observations across 178 entities over 1990 to 2014 in the local executable panel (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Firm-year publication, patenting, and corporate NPL citation files already merged into the executable science panel. External public-research citation counts and internal science-channel counts from the local DISCERN-linked firm-year data. Compustat-linked scale and performance controls already available in the workspace. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
DISCERN-linked firm-year science panel	define the firm-year sample and publication stock
Processed patent and nonpatent citation panel	measure external public-research citations and internal science-channel citations
Compustat-linked firm controls	supply scale, profitability, and R&D controls

Local data families used in the present paper.

3.2 Citation-Channel Measures and Derivation

The construct Public-research knowledge flow is operationalized locally as External public-research citation count and appears in the executable dataset as corp_exter_cites; log_external_public_research. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed external corporate public-research citation count and log-transform it.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Alternative citation channel is operationalized locally as Internal science citation count and appears in the executable dataset as inter_cites; log_internal_science_channel. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed internal science-channel citation count and log-transform it.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files

used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Science openness is operationalized locally as Publication stock and appears in the executable dataset as pub_stock_permno_adj; log_pub_stock. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the already merged publication stock variable.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Subsequent innovation is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Lead the annual patent count by one year within firm.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm controls is operationalized locally as Assets, R&D intensity,

ROA and appears in the executable dataset as log_assets; rd_intensity; roa. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the firm-year controls already present in the science panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Public-research knowledge flow	External public-research citation count	Use the processed external corporate public-research citation count and log-transform it.
Alternative citation channel	Internal science citation count	Use the processed internal science-channel citation count and log-transform it.
Science openness	Publication stock	Carry forward the already merged publication stock variable.
Subsequent innovation	Next-year patent count	Lead the annual patent count by one year within firm.
Firm controls	Assets, R&D intensity, ROA	Reuse the firm-year controls already present in the science panel.

Variable mapping and derivation logic in the main text.

3.3 Executable Sample Reconstruction

The local analysis file contains 2,511 observations drawn from 178 entities. The observed time span runs from 1990 to 2014. Selected sample moments are Next Patents = 222.3, External Public Research = 45.39, Internal Science Channel = 18.89, Publication Stock = 914.7 (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor

the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	2,511
Unique Firms	178
Start Year	1990
End Year	2014
Mean Next Patents	222
Mean External Public Research	45.39
Mean Internal Science Channel	18.89
Mean Publication Stock	914.7

Sample reconstruction from the current local dataset.

3.4 Construct-Level Mapping to the Local Panel

The derivation appendix records how each major construct enters the executable estimand for Roach and Cohen (2013). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Public-research knowledge flow, the manuscript uses External public-research citation count under the exact local implementation corp_exter_cites; log_external_public_research. The derivation step is: Use the processed external corporate public-research citation count and log-transform it.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Alternative citation channel, the manuscript uses Internal science citation count under the exact local implementation inter_cites; log_internal_science_channel. The derivation step is: Use the processed internal science-channel citation count and log-transform it.. This construct is retained because it occupies the same an-

alytic position in the local design that the published paper assigns to the original concept. For Science openness, the manuscript uses Publication stock under the exact local implementation pub_stock_permno_adj; log_pub_stock. The derivation step is: Carry forward the already merged publication stock variable.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Subsequent innovation, the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Lead the annual patent count by one year within firm.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm controls, the manuscript uses Assets, R&D intensity, ROA under the exact local implementation log_assets; rd_intensity; roa. The derivation step is: Reuse the firm-year controls already present in the science panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Evidence on Citation Channels

Because the original article is a measurement paper, the descriptive analysis starts with the science-facing citation channels themselves rather than with a generic innovation profile. For Next-year patents, the mean is 222.3, the standard deviation is 507.6, and the interquartile range runs from 20.00 to 218.2; For External public-research citations, the mean is 45.39, the standard deviation is 173.3, and the interquartile range runs from 0.00000 to 18.00; For Internal science citations, the mean is 18.89, the standard deviation is 86.08, and the interquartile range runs from 0.00000 to 7.000; For Log external public-research citations, the mean is 1.653, the standard deviation is 1.888, and the interquartile range runs from 0.00000 to 2.944 . The main descriptive issue is whether the external public-research channel, the internal science channel, and publication stock are empirically distinct enough to support a real validation-style comparison inside the local panel (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile is read as a science-engagement history. If public-research exposure rises only because the whole panel grows mechanically, the later regression would say less

about measurement validity than the source paper intends. the value of firm-year observations is 133 in 1990 and 75 in 2014; the value of mean next-year patents is 117.6 in 1990 and 462.9 in 2014; the value of mean external public-research citations is 4.887 in 1990 and 151.6 in 2014 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is interpreted as a measurement gradient, asking whether firms with stronger observable public-research exposure also sit in meaningfully different regions of subsequent patenting support. Mean outcomes range from 62.14 to 684.7 across the support of the focal regressor, while the mean regressor itself ranges from 0.359 to 4.757. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation discussion is framed as a validity screen. The question is not merely whether the regressors co-move, but whether the public-research channel still contains information once publication stock, internal science use, and firm scale are visible. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Measurement Reliability and Coverage

For this paper, data quality is fundamentally about measurement reliability. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). Coverage is reported to show whether the science-facing comparison is learned from a broad firm-year panel rather than from a few unusually science-intensive firms. The most heavily represented entities contribute between 25 and 25 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm

names happen to be listed in a long table (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). Support matters here because the validation-style argument only works if firms with different public-research exposure levels are genuinely different empirical cases. In the lowest observed quantile, the mean regressor is 0.359 and the mean outcome is 62.14 across 1,546 observations. In the highest observed quantile, the corresponding values are 4.757, 684.7, and 501 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). These diagnostics matter because a measurement paper can look persuasive while still relying on thin support or channel redundancy. The point of this section is to show that the local science-facing comparison has usable variation, usable coverage, and usable support before the main coefficients are interpreted (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005).

Period	FirmYears	MeanNextPatents	MeanExternalPublications	MeanJournalScience	MeanPublicationStock
1990	133	118	4.887	3.632	634.9
1991	132	118	5.985	3.765	647.1
1992	130	122	6.815	3.862	669.7
1993	126	130	8.437	5.333	694.2
1994	123	129	10.28	7.187	746.9
1995	122	140	11.18	7.098	796.7
1996	123	130	15.11	7.236	668.8
1997	120	170	22.06	6.717	686.0
1998	113	183	29.81	10.60	729.8
1999	106	195	33.22	10.94	774.0
2000	102	209	41.23	10.75	877.0
2001	100	219	48.55	13.85	918.6
2002	98	226	50.70	12.94	896.1
2003	100	216	52.84	14.30	938.6
2004	98	195	48.67	15.09	963.5
2005	93	250	48.32	15.74	1,015
2006	89	232	70.90	25.66	1,082
2007	83	258	72.70	27.89	1,199
2008	78	292	71.04	29.47	1,175
2009	75	387	78.93	34.48	1,337
2010	75	374	99.63	43.40	1,321
2011	72	425	106.9	51.06	1,375

Period	FirmYears	MeanNextPatents	MeanExternalPublicResearch	MeanInternalScience	MeanPublicationStock
2012	70	463	115.1	58.30	1,411
2013	75	468	127.9	64.64	1,310
2014	75	463	151.6	80.07	1,327

Observed time patterns relevant to the focal empirical setting.

Variable	Next-year patents	External public-research citations	Internal science citations	Log external public-research citations	Log internal science citations	Publication stock	Log publication stock	Log assets	R&D intensity	ROA
Next-year patents	1	0.813	0.824	0.558	0.599	0.592	0.429	0.436	0.114	0.067
External public-research citations	1	1.000	0.831	0.576	0.560	0.663	0.367	0.305	0.137	0.078
Internal science citations	1	0.831	1.000	0.449	0.550	0.553	0.307	0.266	0.144	0.105
Log external public-research citations	1	0.576	0.449	1.000	0.876	0.706	0.804	0.561	0.381	0.143
Log internal science citations	1	0.560	0.550	0.876	1.000	0.708	0.727	0.541	0.366	0.152
Publication stock	1	0.663	0.553	0.706	0.708	1.000	0.621	0.469	0.221	0.077

Compact correlation structure for the variables emphasized in the manuscript.

4.2 Science-Facing Channel Patterns

The figure is built as a measurement figure rather than as a generic trend chart (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The left panel tracks the observable science-facing channels that compete for validity in the local panel, while the right panel shows whether subsequent patenting shifts across the public-research exposure gradient. mean external public-research citations moves from 4.887 in 1990 to 151.6 in

2014; mean internal science citations moves from 3.632 in 1990 to 80.07 in 2014; mean publication stock moves from 634.9 in 1990 to 1,327 in 2014 Across the support panel, mean outcome moves from 62.14 to 684.7 as the mean regressor shifts from 0.359 to 4.757 (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

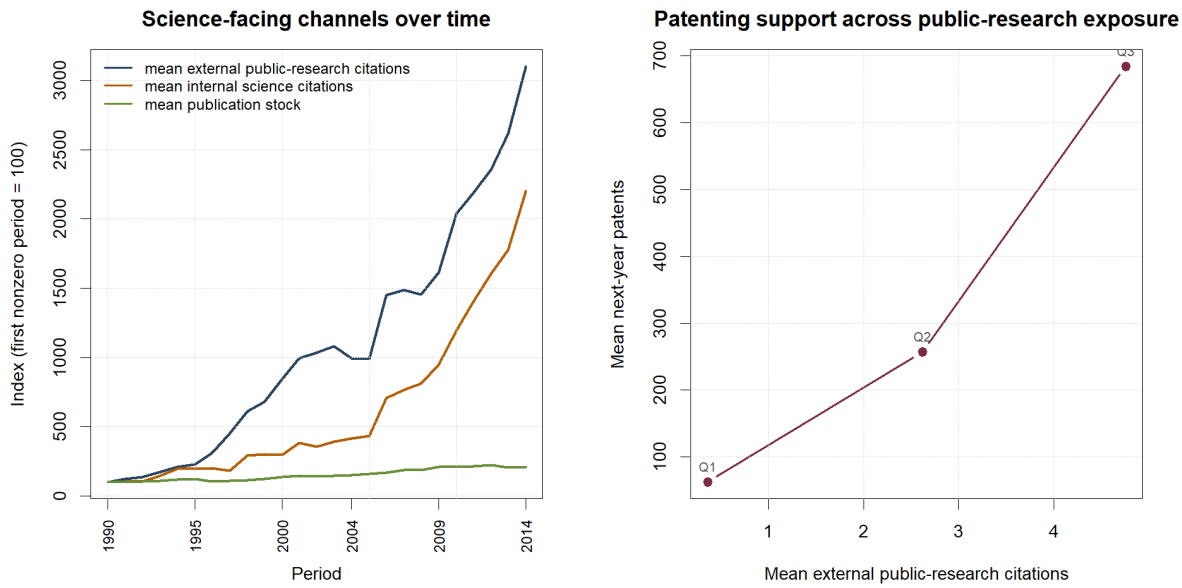

Figure 1: Figure 1. Science-facing citation channels in the local firm-year panel and the resulting public-research exposure gradient.

5 Empirical Design for the Measurement Analogue

The estimand targeted in this manuscript is A firm and year fixed-effects count model of next-year patenting on external public-research citations, internal science citations, publication stock, and firm controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the

inferential claim narrower and more honest (Roach and Cohen, 2013; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Why the Local Substitutions Are Necessary

The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is DISCERN-linked public-firm year panel with publications, patenting, corporate NPL citation counts, and financial controls, and the empirical window is 1990 to 2014 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Next-year patent count at the firm-year level, and the main explanatory design is Firm and year fixed-effects count model relating next-year patenting to external public-research citations, internal science citations, and publication stock. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Publication stock, assets, R&D intensity, and ROA plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps firm-years with verified scale and R&D information and uses only the already merged science panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a

source-data replay (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame DISCERN-linked public-firm year panel with publications, patenting, corporate NPL citation counts, and financial controls over 1990 to 2014 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Next-year patent count at the firm-year level and the main explanatory design is Firm and year fixed-effects count model relating next-year patenting to external public-research citations, internal science citations, and publication stock. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Publication stock, assets, R&D intensity, and ROA plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps firm-years with verified scale and R&D information and uses only the already merged science panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation for the Local Validation Test

$$E[\text{NextPatents}_{it+1} \mid X_{it}] = \exp\left(\beta_1 \log(1 + \text{ExternalPublicResearch}_{it}) + \beta_2 \log(1 + \text{InternalScience}_{it}) + \beta_3 \log(1 + \text{PublicationStock}_{it}) + \beta_4 X_{it} + \alpha_i + \gamma_t\right). \tag{1}$$

6 Measurement Results in the Local Panel

6.1 Channel-Level Estimates

The central local estimate is 0.0220 on External Public-Research Citations (p = 0.59). The local panel treats external public-research citations as the clearest executable public-research channel and compares them directly against an internal citation channel. Internal science citations is estimated

at 0.169 with p = ¡ .001. Log publication stock is estimated at 0.230 with p = ¡ .001. ROA is estimated at 1.455 with p = 0.010 (Roach and Cohen, 2013; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
External public-research citations	0.0220	0.0408	0.589
Internal science citations	0.1692	0.0390	1.43e-05
Log publication stock	0.2300	0.0689	0.0008
Log assets	0.2992	0.1242	0.0160
R&D intensity	1.256	1.408	0.372
ROA	1.455	0.5668	0.0103

Key coefficient table used in the main text.

6.2 Interpreting the Channel Contrast

The coefficient pattern is not being summarized as a bare sign check. Internal science citations is estimated at 0.169 with a p-value of ¡ .001; Log publication stock is estimated at 0.230 with a p-value of ¡ .001; ROA is estimated at 1.455 with a p-value of 0.010; Log assets is estimated at 0.299 with a p-value of 0.016 (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Internal science citations, which is estimated at 0.169 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 222.3 and standard deviation 507.6 in the local panel, while the main regressor Log external public-research citations has mean 1.653 and standard deviation

1.888 (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 62.14 to 684.7 while the mean regressor ranges from 0.359 to 4.757. This does not substitute for the model, but it shows how much raw variation the specification must explain (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local panel treats external public-research citations as the clearest executable public-research channel and compares them directly against an internal citation channel. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Comparison Across Citation Channels

The coefficient figure helps the reader compare the external public-research channel directly with the internal science channel and publication stock without relying on a dense coefficient table alone (Roach and Cohen, 2013; Hall et al., 2005).

6.4 Specification Comparisons

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Internal science citations with estimate 0.169 and p = ¡ .001 (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Internal science citations is estimated at 0.169 with p = ¡ .001 and is classified as a strong signal; Log publication stock is estimated at 0.230 with p = ¡ .001 and is classified as a strong signal; ROA is estimated at 1.455 with p = 0.010 and is classified as a moderate signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another,

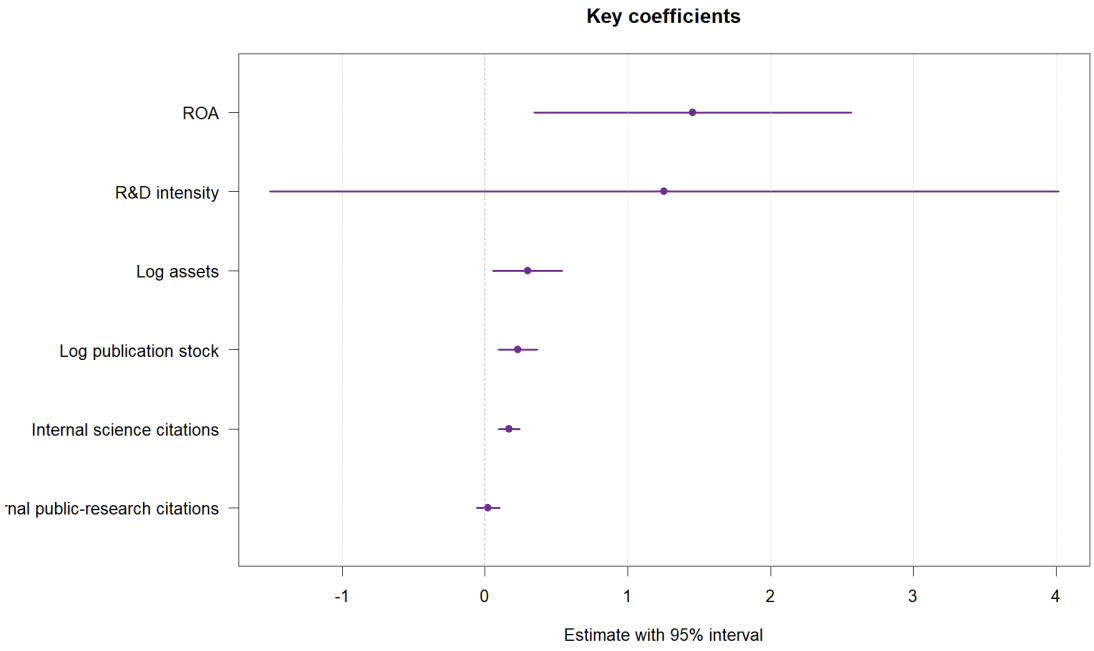

Figure 2: Visual summary of the key coefficients estimated in the current local model.

and the paper should show that pattern directly rather than compress it into a single headline coefficient (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Roach and Cohen (2013) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.5 How the Coefficient Families Line Up

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Roach

and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Internal science citations = 0.169 (p = ¡ .001); Log publication stock = 0.230 (p = ¡ .001); ROA = 1.455 (p = 0.010); Log assets = 0.299 (p = 0.016); R&D intensity = 1.256 (p = 0.37). The largest positive estimate in this specification is ROA at 1.455, while the largest negative estimate is External public-research citations at 0.0220. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Roach and Cohen (2013) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Roach and Cohen, 2013; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.6 Model Diagnostics

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). For public_research_channels, the current fit export reports N = 2,408, pseudo R-squared = 0.931, R-squared = NA (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Roach and Cohen (2013) is being carried out on a clearly described estimating sample with observable fit statistics (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005).

7 Comparison with the Published Validation Evidence

7.1 Published Validation Logic Versus the Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Roach and Cohen, 2013). For Roach and Cohen (2013), the local headline result is External Public-Research Citations = 0.0220 with p = 0.59. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper finds that public-research knowledge flows are measured more cleanly by nonpatent references than by patent references.	External Public-Research Citations = 0.0220 with p = 0.59.	The published paper argues that nonpatent references are better indicators of knowledge flows from public research than patent references. The local analogue asks a narrower but compatible question: when external public-research citations and internal science-channel citations are observed together, which channel is more strongly associated with subsequent patenting in the current firm-year panel?

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published design compares survey-based knowledge-flow measures with backward patent and nonpatent reference channels.	The local analogue uses a firm-year panel with external public-research citation counts, internal science-channel citations, and publication stocks to evaluate which observable channel is most strongly tied to subsequent patenting.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample matches patents to research-and-development labs and survey-based measures of knowledge flows from public research.	DISCERN-linked public-firm year panel with publications, patenting, corporate NPL citation counts, and financial controls. Local sample = 2,511 observations across 178 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 What the Comparison Means

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design compares survey-based knowledge-flow measures with backward patent and non-patent reference channels By contrast, the local paper relies on the following executable design: The local analogue uses a firm-year panel with external public-research citation counts, internal science-

channel citations, and publication stocks to evaluate which observable channel is most strongly tied to subsequent patenting These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published sample matches patents to research-and-development labs and survey-based measures of knowledge flows from public research The local sample instead works from DISCERN-linked public-firm year panel with publications, patenting, corporate NPL citation counts, and financial controls and yields 2,511 observations across 178 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that nonpatent references are better indicators of knowledge flows from public research than patent references. The local analogue asks a narrower but compatible question: when external public-research citations and internal science-channel citations are observed together, which channel is more strongly associated with subsequent patenting in the current firm-year panel? The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The published paper argues that nonpatent references are better indicators of knowledge flows from public research than patent references. The local analogue asks a narrower but compatible question: when external

public-research citations and internal science-channel citations are observed together, which channel is more strongly associated with subsequent patenting in the current firm-year panel? The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The published paper argues that nonpatent references are better indicators of knowledge flows from public research than patent references. The local analogue asks a narrower but compatible question: when external public-research citations and internal science-channel citations are observed together, which channel is more strongly associated with subsequent patenting in the current firm-year panel? (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Roach and Cohen, 2013; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The published design uses survey-linked lab-level evidence to validate citation measures directly (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun works with processed firm-year science and citation channels rather than a survey match to observed knowledge

flows (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The paper therefore tests whether the relative empirical value of public-research-oriented citation channels survives in the executable local panel (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent local measurement analogue that compares public-research-oriented citation channels inside a current executable firm-year panel (Roach and Cohen, 2013; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Roach and Cohen, 2013; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	2,511	rows
Unique Firms	178	unique_firms
Start Year	1990	year_min
End Year	2014	year_max
Mean Next Patents	222	mean_next_patents
Mean External Public Research	45.39	mean_external_public_research
Mean Internal Science Channel	18.89	mean_internal_science_channel
Mean Publication Stock	914.7	mean_publication_stock

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Next-year patents	2,511	222.3	507.6	20.00	76.00	218.2	0	7662
External public-research citations	2,511	45.39	173.3	0.0000	2.000	18.00	0	2721
Internal science citations	2,511	18.89	86.08	0.0000	0.0000	7.000	0	1914
Log external public-research citations	2,511	1.653	1.888	0.0000	1.099	2.944	0	8
Log internal science citations	2,511	1.137	1.588	0.0000	0.0000	2.079	0	8
Publication stock	2,511	914.7	1,920	24.93	147.1	964.3	0	15607
Log publication stock	2,511	4.926	2.352	3.255	4.998	6.872	0	10
Log assets	2,511	9.345	1.353	8.445	9.271	10.23	4	14
R&D intensity	2,511	0.0415	0.0400	0.0132	0.0280	0.0593	0	1
ROA	2,511	0.0621	0.0751	0.0266	0.0631	0.0980	-1	1

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Next-year patents	External public-research citations	Internal science citations	Log external public-research citations	Log internal science citations	Publication stock	Log publication stock	Log assets	R&D intensity	ROA
Next-year patents	1	0.813	0.824	0.558	0.599	0.592	0.429	0.436	0.114	0.067
External public-research citations	1	1.000	0.831	0.576	0.560	0.663	0.367	0.305	0.137	0.078
Internal science citations	1	0.831	1.000	0.449	0.550	0.553	0.307	0.266	0.144	0.105
Log external public-research citations	1	0.576	0.449	1.000	0.876	0.706	0.804	0.561	0.381	0.143
Log internal science citations	1	0.560	0.550	0.876	1.000	0.708	0.727	0.541	0.366	0.152
Publication stock	1	0.663	0.553	0.706	0.708	1.000	0.621	0.469	0.221	0.077
Log publication stock	0	0.367	0.307	0.804	0.727	0.621	1.000	0.556	0.327	0.133
Log assets	0	0.305	0.266	0.561	0.541	0.469	0.556	1.000	-0.133	0.034
R&D intensity	0	0.137	0.144	0.381	0.366	0.221	0.327	-0.133	1.000	0.181
ROA	0	0.078	0.105	0.143	0.152	0.077	0.133	0.034	0.181	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanNextPatents	MeanExternalPublications	MeanInternalScience	MeanPublicationStock
1990	133	118	4.887	3.632	634.9
1991	132	118	5.985	3.765	647.1
1992	130	122	6.815	3.862	669.7
1993	126	130	8.437	5.333	694.2
1994	123	129	10.28	7.187	746.9
1995	122	140	11.18	7.098	796.7
1996	123	130	15.11	7.236	668.8
1997	120	170	22.06	6.717	686.0
1998	113	183	29.81	10.60	729.8
1999	106	195	33.22	10.94	774.0
2000	102	209	41.23	10.75	877.0
2001	100	219	48.55	13.85	918.6
2002	98	226	50.70	12.94	896.1
2003	100	216	52.84	14.30	938.6
2004	98	195	48.67	15.09	963.5
2005	93	250	48.32	15.74	1,015
2006	89	232	70.90	25.66	1,082
2007	83	258	72.70	27.89	1,199
2008	78	292	71.04	29.47	1,175
2009	75	387	78.93	34.48	1,337
2010	75	374	99.63	43.40	1,321
2011	72	425	106.9	51.06	1,375
2012	70	463	115.1	58.30	1,411
2013	75	468	127.9	64.64	1,310
2014	75	463	151.6	80.07	1,327

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	1,546	0.359	62.14
Q2	464	2.617	256.9
Q3	501	4.757	684.7

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
log_external_public_research	0.0220	0.0408	0.540	0.59
log_internal_science_channel	0.1692	0.0390	4.340	¡ .001
log_pub_stock	0.2300	0.0689	3.336	¡ .001
log_assets	0.2992	0.1242	2.408	0.016
rd_intensity	1.256	1.408	0.892	0.37
roa	1.455	0.5668	2.567	0.010

Full coefficient dump exported from the local estimation script.

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
public_research_channels	2,408	-39,433	79,243	80,330	0.931	NA	NA

Model fit statistics from the local estimation output.

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